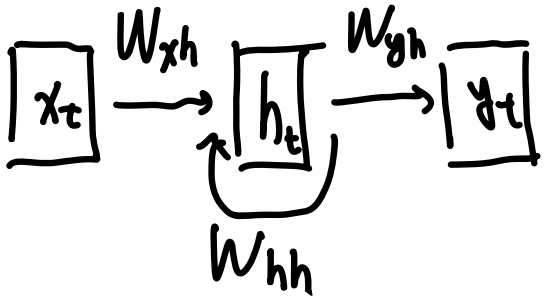
Recap:

① NPLM:

$$\begin{cases} \hat{p}(w_t \mid w_{t-N+1:t-1}) = \dfrac{e^{y_{w_t}}}{\sum_i e^{y_i}} \\ y = b + Wx + U \cdot \tanh(d + Hx) \\ x = (c_{t-N+1}, \cdots, c_{t-1}) \\ \theta = (b, d, W, U, H, C) \end{cases}$$

② RNN:



$$\begin{cases} \hat{p}(w_t \mid w_{1:t-1}) = \hat{y}_t[w_t] \\ \hat{y}_t = \text{softmax}(o_t) = \phi_o(o_t) \\ o_t = W_{yh} \cdot h_t + b_y \\ h_t = \phi_h(W_{xh}^T x_t + W_{hh}^T h_{t-1} + b_h), \ \phi_h: \tanh / \text{ReLU} \end{cases}$$

$$L(\hat{y}, y) = -\sum_{t=1}^{T} L_t(\hat{y}_t, y_t)$$, BPTT:

$$\frac{\partial L}{\partial W_{yh}} = \sum_{t=1}^{T} \frac{\partial L_t}{\partial \hat{y}_t} \cdot \frac{\partial \hat{y}_t}{\partial o_t} \cdot \frac{\partial o_t}{\partial W_{yh}}$$

$$\frac{\partial L}{\partial W_{hh}} = \sum_{t=1}^{T} \sum_{k=1}^{t} \frac{\partial L_t}{\partial \hat{y}_t} \cdot \frac{\partial \hat{y}_t}{\partial h_t} \cdot \frac{\partial h_t}{\partial h_k} \cdot \frac{\partial h_k}{\partial W_{hh}}$$

$$\frac{\partial h_t}{\partial h_k} = \prod_{j=k}^{t-1} \frac{\partial h_{j+1}}{\partial h_j}$$

$$\frac{\partial L}{\partial W_{xh}} = \sum_{t=1}^{T} \sum_{k=1}^{t} \frac{\partial L_t}{\partial \hat{y}_t} \cdot \frac{\partial \hat{y}_t}{\partial h_t} \cdot \frac{\partial h_t}{\partial h_k} \cdot \frac{\partial h_k}{\partial W_{xh}}$$

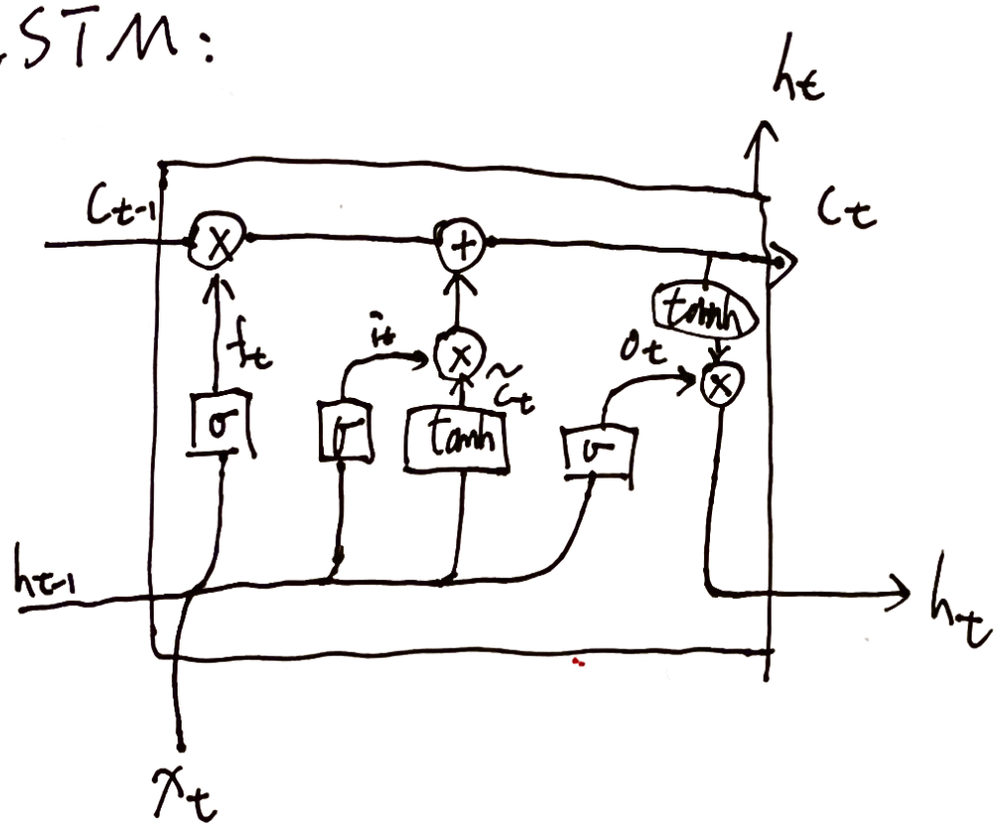
- Sequential style!

$$\left\| \frac{\partial h_t}{\partial h_k} \right\| \leq \prod_{i=k}^{t-1} \left\| \frac{\partial h_{i+1}}{\partial h_i} \right\|$$

$$\lesssim \left(\| \mathrm{Diag}(\phi_h') \cdot \| W_{hh} \| \right)^{t-k}$$

$$\Rightarrow \begin{cases} 0 & \text{vanishing gradients} \\ \infty. & \text{exploding gradients} \end{cases}$$

LSTM:



① Cell state $C_t = f_t \cdot C_{t-1} + i_t \cdot \tilde{C}_t$

② forget some info: $f_t = \sigma(W_f \cdot [h_{t-1}, x_t] + b_f)$.

↓

③ store new info: $\begin{cases} \text{input gate layer: which values will be updated.} \\ \text{tanh layer: } \tilde{C}_t \text{ could be added into } C_t \end{cases}$

↓

↳ replace the old ones we forget:

$$i_t = \sigma(W_i \cdot [h_{t-1}, x_t] + b_i)$$

$$\tilde{C}_t = \tanh(W_c \cdot [h_{t-1}, x_t] + b_c).$$

↓

cell state update: $C_t = f_t \cdot C_{t-1} + i_t \cdot \tilde{C}_t$.

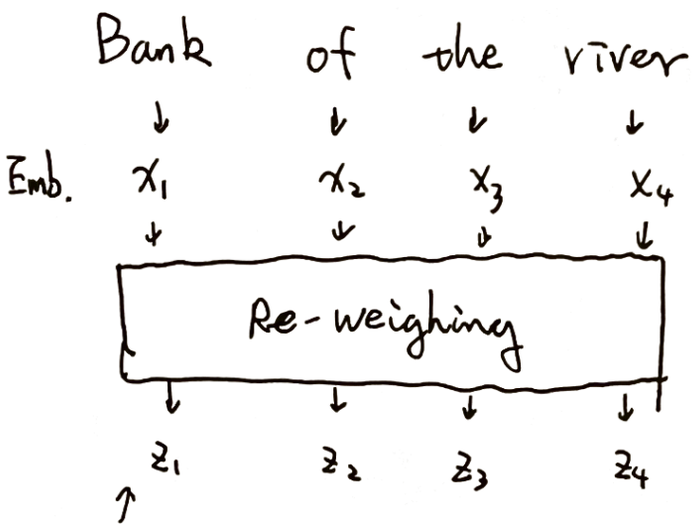
④. Output gate: $\underbrace{\text{filtered cell states}}_{\sigma(h_{t-1})}$ and olde info.

$$o_t = \sigma(W_o \cdot [h_{t-1}, x_t] + b_o)$$

$$h_t = o_t \cdot \tanh(C_t)$$

① Attention via Reweighing:

$x_i \in \mathbb{R}^d, \quad z_i \in \mathbb{R}^d$



better and more context info.

$$x_1 \cdot x_1 = \tilde{w}_{11}$$
$$x_1 \cdot x_2 = \tilde{w}_{12}$$
$$x_1 \cdot x_3 = \tilde{w}_{13}$$
$$x_1 \cdot x_4 = \tilde{w}_{14}$$

$\xrightarrow{\text{Softmax}}$ w_{11}, w_{12}, w_{13}, w_{14} $\quad \sum_{i=1}^{4} w_{1i} = 1$

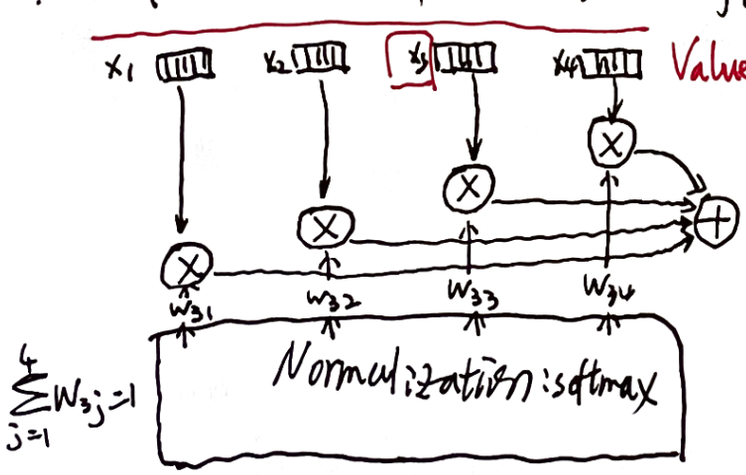
$$z_1 = \sum_{i=1}^{4} w_{1i} \cdot x_i$$

better context representation of x_1

$\downarrow$ make this idea as model learning $\rightarrow$ add some weights

② Self-attention mechanism: $f_\theta : \mathbb{R}^{n \times d} \mapsto \mathbb{R}^{n \times d}$, $f_\theta(x) = z$.

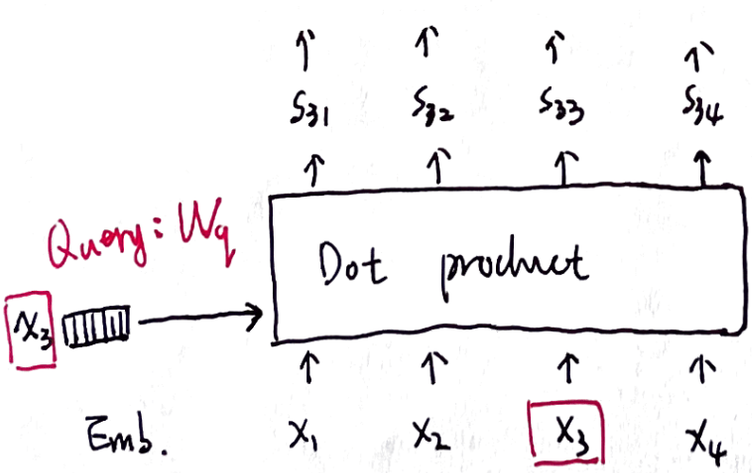
Value: $W_v \in \mathbb{R}^{d \times d_v}$



$$z_3 = \sum_{j=1}^{4} w_{3j} \cdot x_j$$

$\sum_{j=1}^{4} w_{3j} = 1$

x_3 is used in $\begin{cases} \text{Querys} \\ \text{Keys} \\ \text{Values} \end{cases} \rightarrow z_3$



Key: $W_k \in \mathbb{R}^{d \times d_k}$

$$\begin{cases} Q(x_i) = W_q^T \cdot x_i, & W_q \in \mathbb{R}^{d \times d_k} \\ K(x_i) = W_k^T \cdot x_i, & W_k \in \mathbb{R}^{d \times d_k} \\ V(x_i) = W_v^T \cdot x_i, & W_v \in \mathbb{R}^{d \times d_v} \end{cases}$$

$\downarrow$

$$\text{Attention}(Q, K, V) = \text{softmax}\left(\frac{QK^T}{\sqrt{d_k}}\right) \cdot V$$

$$\theta = [W_q, W_k, W_v].$$

③. <u>Multi-head attention</u>. $f_\theta : \mathbb{R}^{n\times d} \mapsto \mathbb{R}^{n\times d}$, $f_\theta(x) = z$

To make model more powerful, add more parameters.

- Multiple self-attention : $h = 1, 2, \cdots, H$, $H = 8$.

$\theta' = [W_q, W_k, W_v] \rightarrow \theta_h = [W_{h,q}, W_{h,k}, W_{h,v}]$, $h = 1, 2, \cdots, H$.

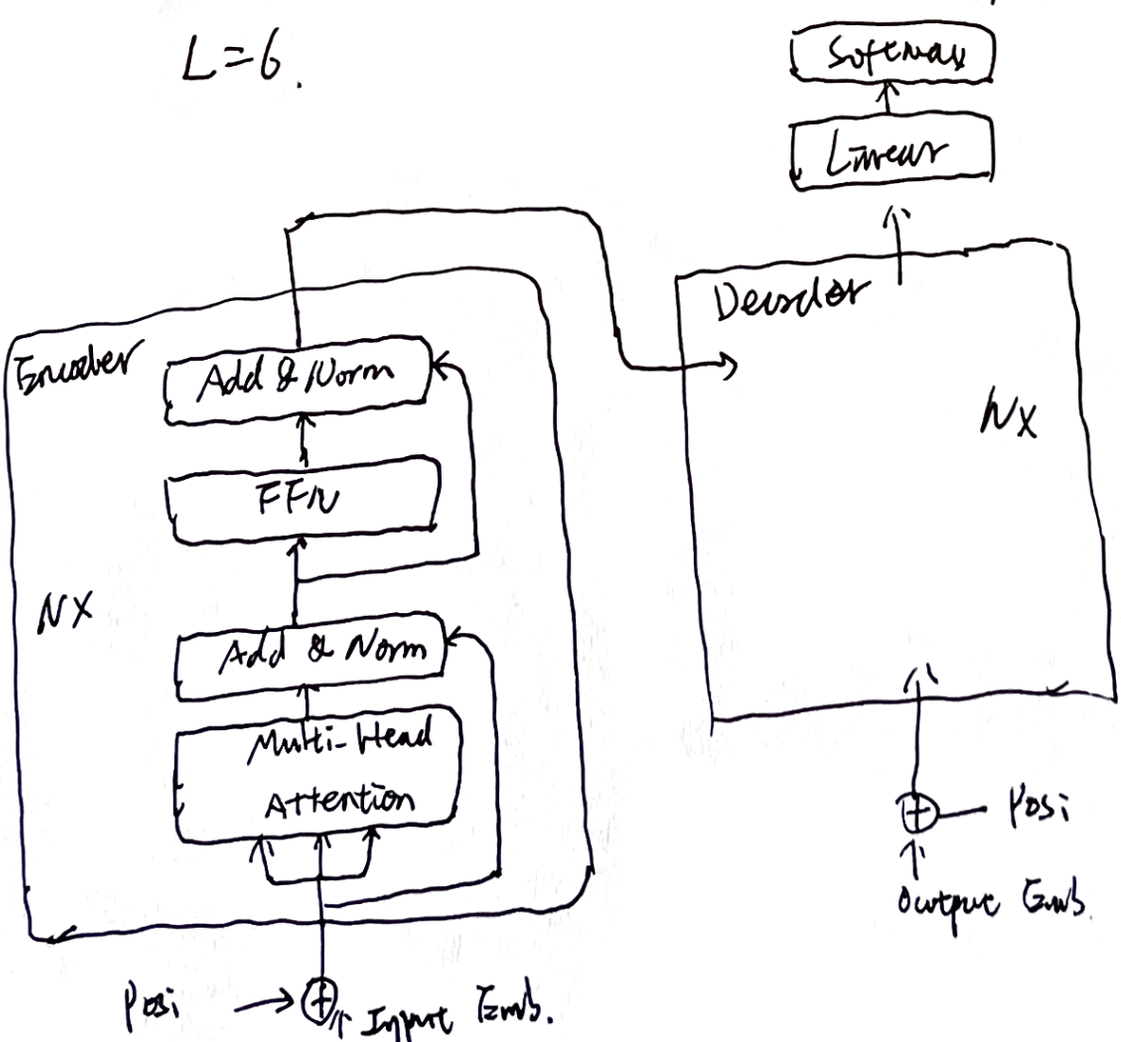
$$\text{Multihead}(Q, K, V) = \text{Concat}(\text{head}_1, \text{head}_2, \cdots, \text{head}_h) \cdot W^o$$

$$\text{head}_i = \text{Attention}(Q_i, K_i, V_i).$$

④. <u>Transformer</u>. $f_\theta : \mathbb{R}^{n\times d} \mapsto \mathbb{R}^{n\times d}$, $f_\theta(x) = z$.

- Encoder : $f_{\theta_L^E} \circ f_{\theta_{L-1}^E} \circ \cdots f_{\theta_2^E} \circ f_{\theta_1^E}(x) \in \mathbb{R}^{n\times d}$.
- Decoder : $f_{\theta_L^D} \circ f_{\theta_{L-1}^D} \circ \cdots f_{\theta_2^D} \circ f_{\theta_1^D}(x) \in \mathbb{R}^{n\times d}$.

$L = 6$.



Transformer in math:

A $\boxed{\text{transformer block}}$ is a parameterized function class

$$f_\theta : \mathbb{R}^{n\times d} \longmapsto \mathbb{R}^{n\times d}.$$

If $x \in \mathbb{R}^{n\times d}$ → collection of n objects, then $f_\theta(x) = z$ where

Multi Head

$$Q^{(h)}(x_i) = W_{h,q}^T \cdot x_i \qquad W_{h,q} \in \mathbb{R}^{d\times k}$$

$$K^{(h)}(x_i) = W_{h,k}^T \cdot x_i, \qquad W_{h,k} \in \mathbb{R}^{d\times k}$$

$$V^{(h)}(x_i) = W_{h,v}^T \cdot x_i \qquad W_{h,v} \in \mathbb{R}^{d\times k}.$$

$$\alpha_{i,j}^{(h)} = \mathrm{softmax}_j\left(\frac{\langle Q^{(h)}(x_i), K^{(h)}(x_j)\rangle}{\sqrt{k}}\right),$$

$$u_i' = \sum_{h=1}^{H} W_{c,h}^T \sum_{j=1}^{n} \alpha_{i,j}^{(h)} V^{(h)}(x_j), \qquad W_{c,h} \in \mathbb{R}^{k\times d}$$

Add & Norm

$$u_i = \mathrm{LayerNorm}(x_i + u_i' ; \gamma_1, \beta_1) \qquad \gamma_1, \beta_1 \in \mathbb{R}^d$$

FFN

$$z_i' = W_2^T \cdot \mathrm{ReLU}(W_1^T \cdot u_i) \qquad W_1 \in \mathbb{R}^{d\times m},\ W_2 \in \mathbb{R}^{m\times d},$$

Add & Norm

$$z_i = \mathrm{LayerNorm}(u_i + z_i' ; \gamma_2, \beta_2) \qquad \gamma_2, \beta_2 \in \mathbb{R}^d,$$

where LayerNorm is defined for $z \in \mathbb{R}^k$ by

$$\mathrm{LayerNorm}(z ; \gamma, \beta) = \gamma \cdot \frac{(z - \mu_z)}{\sigma_z} + \beta, \qquad \gamma, \beta \in \mathbb{R}^k$$

$$\mu_z = \frac{1}{k}\sum_{i=1}^{k} z_i, \quad \sigma_z = \sqrt{\frac{1}{k}\sum_{i=1}^{k}(z_i - \mu_z)^2},$$

Transformer: $f_{\theta_L} \circ f_{\theta_{L-1}} \circ \cdots \circ f_{\theta_2} \circ f_{\theta_1}(x) \in \mathbb{R}^{n\times d}$, $\theta_i = \begin{cases} W_{h,q}, W_{h,k}, W_{h,v} \\ W_{c,h}, \gamma_1, \beta_1 \\ W_1, W_2, \gamma_2, \beta_2 \end{cases}$

$f_\theta = f_{[\theta_L, \cdots \theta_1]}$,

hyper-parameters of the Transformer:

- d, k, m, H, and L
- $d := d_{model} = 512$: defines the dimensionality of input and output.
- $k := d_k = d_v = d_{model} / H = 64$: self-attention size
- H : The number of heads used in each multi-head attention
  $H = 8$ in the Transformer model.
- $m := d_{ff} = 2048$: feed-forward network size.
- $L := 6$ number of layers for encoder / decoder

Positional Encoding: $PE \in \mathbb{R}^{n \times d}$

$$\begin{cases} PE_{(pos, 2i)} = \sin(pos / 10000^{2i/d}) \\ PE_{(pos, 2i+1)} = \cos(pos / 10000^{2i/d}). \end{cases}$$

$$z = W_{zx}^T \cdot \mathrm{ReLU}(W_x^T \cdot x)$$

Time complexity for 1 layer:

Self-attention: $O(n^2 \cdot d)$. Sequential operations: $O(1)$

Recurrent: $O(n \cdot d^2)$ Sequential operations: $O(n)$.

Input / Output Embedding:

- Encoder: $W_x^T \cdot x + PE$, • Decoder: $W_x^T \cdot x + PE$

Softmax Layer: $P_\theta(y \mid z = f_\theta(x))$ $z := f_\theta(x)_n$.

$$\hat{y} = \mathrm{softmax}(W_z^T \cdot z) = \frac{\exp(W_z^T \cdot z)}{\sum_k \exp(W_z^T \cdot z)_k} \qquad W_z \in \mathbb{R}^{d \times o}.$$

En-De : $|V| = 37{,}000$ model size :

$L := N = 6$, $d := d_{model} = 512$, $m := d_{ff} = 2048$, $h = 8$, $d_k = d_v = 64$.

Total model size = A1 + A2

A1 = Input Emb. + Output Emb. + Softmax layer

A2 = N× Encoder layer + N× Decoder layer

A1 = Weight sharing $W^{emb} \in \mathbb{R}^{|V| \times d_{model}}$ $= 37{,}000 \times 512 = 18{,}944{,}000$ $\approx (18.06)$ MB

Encoder layer = Multihead + FFN + 2· Add & Norm

Decoder layer = 2· Multihead + FFN + 3· Add & Norm

$$\text{Encoder} = 8 \cdot (|W_i^Q| + |W_i^K| + |W_i^V|) + |W_o| + \text{FFN} + 2 \cdot \text{Add \& Norm}$$

$$= \underline{8 \cdot (512 \times 64 \times 3) + 512 \times 512} + \text{FFN} + 2 \cdot 512 \cdot 2$$

$$\text{FFN} = 2 \cdot d_{model} \times d_{ff} = 2 \cdot 512 \times 2048 = 2 \cdot 1024^2.$$

$$\text{Add \& Norm} = 512 \cdot 4 = 2 \cdot 1024$$

$$\text{Encoder} = 4 \cdot 512 \cdot 512 + 2 \cdot 1024^2 + 2 \cdot 1024$$

$$= 3 \cdot 1024^2 + 2 \cdot 1024.$$

$$A2 = 6 \cdot (3 \times 1024^2 + 2 \cdot 1024) + 6 \cdot (4 \times 1024^2 + 3 \times 1024)$$

$$= (18 + 24 + 0.029)\ \text{MB}.$$

$A1 + A2 \approx 61.5$ MB. $\xrightarrow{\times 4}$ needs ≈ 246 MB memory.

The Transformer Model in Equations

John Thickstun

Abstract

This document presents a precise mathematical definition of the transformer model introduced by Vaswani et al. [2017], along with some discussion of the terminology and intuitions commonly associated with the transformer. We also draw some connections between the transformer and lstm, based on observations by Levy et al. [2018].

1 Introduction

A **transformer block** is a parameterized function class $f_\theta : \mathbb{R}^{n\times d} \to \mathbb{R}^{n\times d}$. If $\mathbf{x} \in \mathbb{R}^{n\times d}$ then $f_\theta(\mathbf{x}) = \mathbf{z}$ where

$$Q^{(h)}(\mathbf{x}_i) = W_{h,q}^T \mathbf{x}_i, \quad K^{(h)}(\mathbf{x}_i) = W_{h,k}^T \mathbf{x}_i, \quad V^{(h)}(\mathbf{x}_i) = W_{h,v}^T \mathbf{x}_i, \qquad W_{h,q}, W_{h,k}, W_{h,v} \in \mathbb{R}^{d\times k}, \tag{1}$$

$$\alpha_{i,j}^{(h)} = \mathrm{softmax}_j \left(\frac{\langle Q^{(h)}(\mathbf{x}_i), K^{(h)}(\mathbf{x}_j)\rangle}{\sqrt{k}} \right), \tag{2}$$

$$\mathbf{u}_i' = \sum_{h=1}^{H} W_{c,h}^T \sum_{j=1}^{n} \alpha_{i,j}^{(h)} V^{(h)}(\mathbf{x}_j), \qquad W_{c,h} \in \mathbb{R}^{k\times d}, \tag{3}$$

$$\mathbf{u}_i = \mathrm{LayerNorm}(\mathbf{x}_i + \mathbf{u}_i'; \gamma_1, \beta_1), \qquad \gamma_1, \beta_1 \in \mathbb{R}^d, \tag{4}$$

$$\mathbf{z}_i' = W_2^T \mathrm{ReLU}(W_1^T \mathbf{u}_i), \qquad W_1 \in \mathbb{R}^{d\times m}, W_2 \in \mathbb{R}^{m\times d}, \tag{5}$$

$$\mathbf{z}_i = \mathrm{LayerNorm}(\mathbf{u}_i + \mathbf{z}_i'; \gamma_2, \beta_2), \qquad \gamma_2, \beta_2 \in \mathbb{R}^d. \tag{6}$$

The notation $\mathrm{softmax}_j$ indicates we take the softmax (defined in Equation 9) over the d-dimensional vector indexed by j. The LayerNorm function [Lei Ba et al., 2016] is defined for $\mathbf{z} \in \mathbb{R}^k$ by

$$\mathrm{LayerNorm}(\mathbf{z}; \gamma, \beta) = \gamma \frac{(\mathbf{z} - \mu_\mathbf{z})}{\sigma_\mathbf{z}} + \beta, \qquad \gamma, \beta \in \mathbb{R}^k. \tag{7}$$

$$\mu_\mathbf{z} = \frac{1}{k}\sum_{i=1}^{k} \mathbf{z}_i, \quad \sigma_\mathbf{z} = \sqrt{\frac{1}{k}\sum_{i=1}^{k} (\mathbf{z}_i - \mu_\mathbf{z})^2}. \tag{8}$$

The parameters θ consist of the entries of the weight matrices W, along with the LayerNorm parameters γ and β indicated on the right-hand side. The input $\mathbf{x} \in \mathbb{R}^{n\times d}$ should be interpreted as a collection of n objects, each with d features (often, but not always, a length-n sequence of d-vectors). Observe that the output $\mathbf{z} \in \mathbb{R}^{n\times d}$ has the same structure as the input $\mathbf{x} \in \mathbb{R}^{n\times d}$; a **transformer** is a composition of L transformer blocks, each with their own parameters: $f_{\theta_L} \circ \cdots \circ f_{\theta_1}(\mathbf{x}) \in \mathbb{R}^{n\times d}$. The hyper-parameters of the transformer are d, k, m, H, and L. Common settings of these hyper-parameters are $d = 512$, $k = 64$, $m = 2048$, $H = 8$. The original paper set $L = 6$, but more recent work seems to stack these blocks much deeper.